

BTEC Level 3 Applied Science · Unit 1 Biology · Answer sheet

Prokaryotic and Eukaryotic Cells — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Eukaryotes have a nucleus / membrane-bound organelles; prokaryotes do not.
2. Cell wall, membrane, loop of DNA, ribosomes; also plasmid / flagellum / capsule.
3. Neither — they are acellular / not cells.

B

4. Bacterium: circular loop in cytoplasm, plasmids possible. Animal: linear chromosomes in a nucleus, with histones.
5. Small extra DNA circle; may carry resistance genes or be used as a vector.
6. Bacteria about 1–5 μm ; animal cells about 10–30 μm (accept 10–100 μm).

C

7. Antibiotics can target peptidoglycan walls or 70S ribosomes, which human cells lack or have in a different form.
8. Rapid binary fission plus mutation and selection; plasmids can transfer resistance.
9. Eukaryotic — mitochondria are membrane-bound organelles.

D

10. Nucleus / mitochondria / DNA form / ribosome type / wall / size — any three paired differences.
11. Carry resistance genes; transferable; used as vectors / make infections harder to treat.
12. No nucleus / DNA not enclosed by a nuclear envelope / no membrane-bound organelles.

E

13. These features resemble free-living bacteria, supporting the idea that the organelles originated from engulfed prokaryotes.